

SOLUTION OVERVIEW · FOR SCHOOL LEADERSHIP

ICSC Connect School Management Platform

A centralized system to run attendance, results, examinations, and parent engagement — built around the ICSC curriculum and ready to deploy on your own infrastructure.

Attendance Tracking

ICSC Report Cards

Examination Schedule

Parent Engagement

Multi-role Access

Secure & Auditable

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Document Type: Architecture & Capability Overview

Audience: Principal · Board · IT Committee

1. Executive Summary

What ICSC Connect does, and why it matters for your school.

ICSC Connect is a centralized, web-based platform that brings four critical school operations into a single secure system:

MODULE 01

Student Attendance

Daily roll call by class teachers, automatic % tracking, alerts for students below 75%, leave-request workflow with parent visibility.

MODULE 02

Results & Performance

Subject-wise marks entry, ICSC-compliant report cards with class rank, performance trend analytics, scheduled publication.

MODULE 03

Examinations

Plan exam periods, allocate halls, generate clash-free schedules, print admit cards in bulk.

MODULE 04

Communication

Multi-channel notifications (in-app + email; SMS optional) with a daily personalized digest to every parent.

Why now. ICSC Connect replaces 6+ disconnected spreadsheets, WhatsApp groups, and manual report-card workflows with one auditable platform that scales to thousands of students — without locking you into a vendor cloud.

By the numbers (sample deployment)

6

USER ROLES

4

CORE MODULES

100%

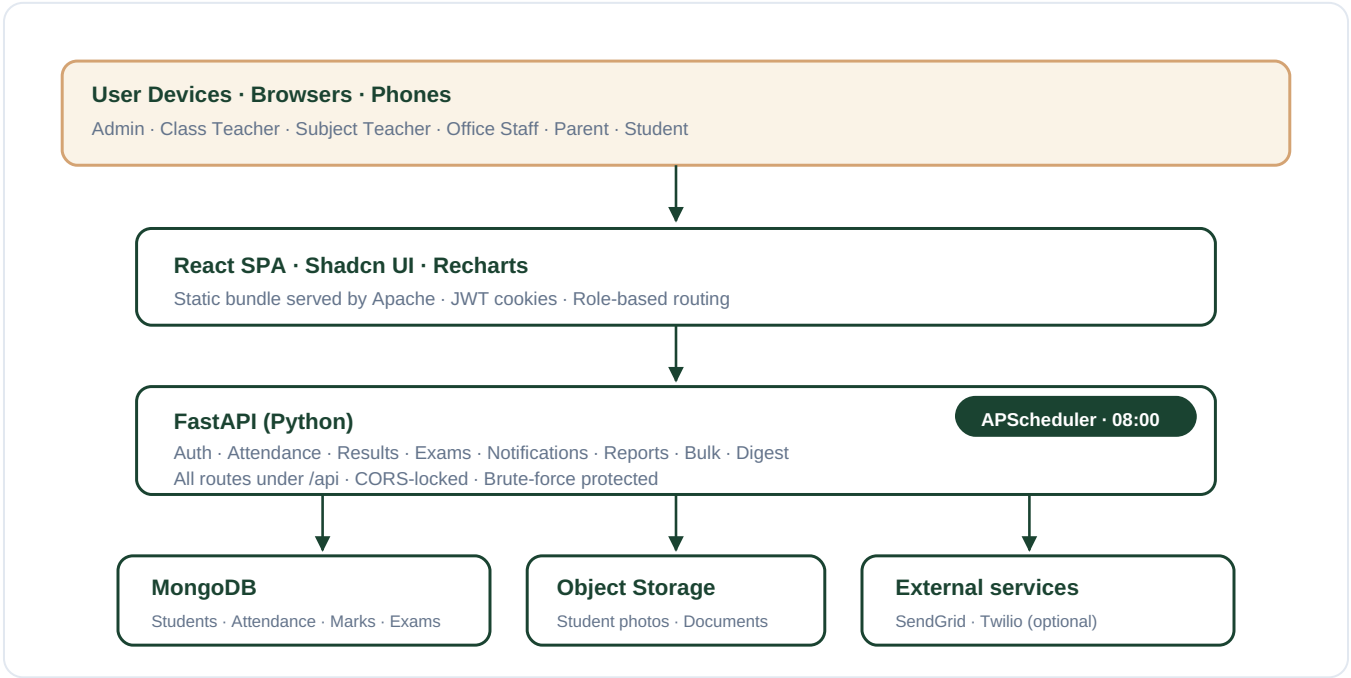
ICSC-ALIGNED

24×7

PARENT ACCESS

2. System Architecture

Modern three-tier architecture that runs on your own server.



Layers explained

Layer	Responsibility
User Devices	Any modern browser (desktop, tablet, phone). No installation required.
Frontend (React)	Single-page app served as static files by Apache. Role-aware navigation, charts, printable report cards and admit cards.
Backend (FastAPI)	Stateless HTTP API. Handles authentication, business rules, report generation, scheduled jobs.
Database (MongoDB)	Stores students, attendance, marks, exams, notifications. Indexed for fast lookups.
Object Storage	Stores student photos and documents separately from the database.
External services	SendGrid for email; Twilio (optional) for SMS. Each can be toggled on or off.

Deployable anywhere. The stack runs on a single Ubuntu server with Apache as the web server. No proprietary cloud is required.

3. User Roles & Permissions

Each role sees exactly what they need — and nothing more.

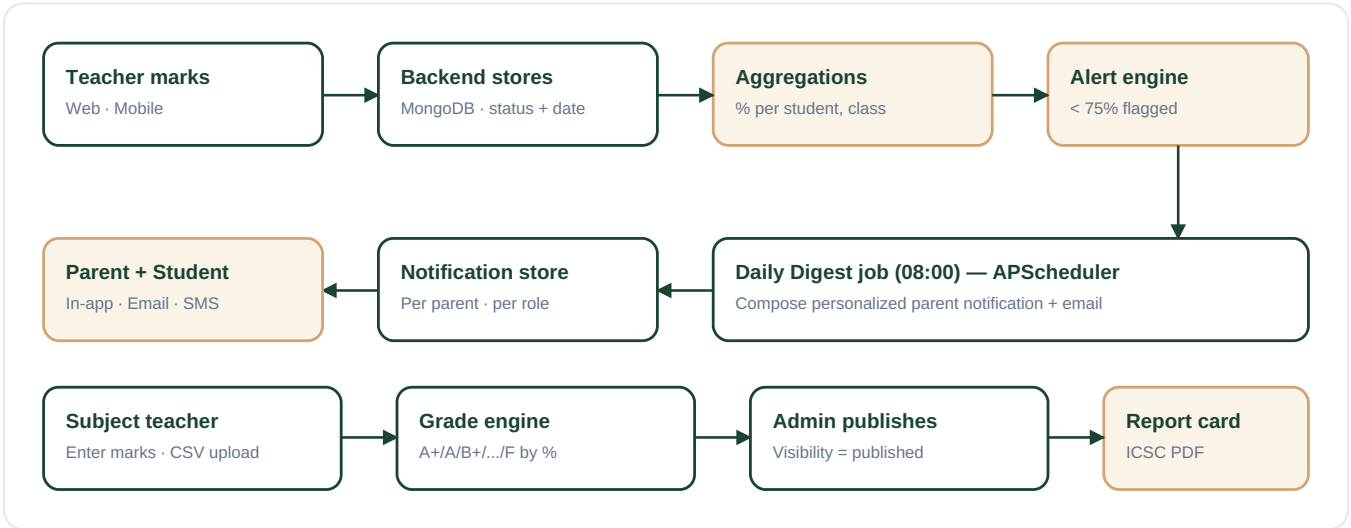
Admin / Principal Full access • Manage students, teachers, classes • Trigger daily digest, publish results • View school-wide reports	Class Teacher Own class + section • Mark daily attendance • Upload student photos • Approve leave requests	Subject Teacher Assigned subjects • Enter marks per exam • Bulk CSV upload • View own class analytics
Office Staff Administrative ops • Exam scheduling, hall planning • Admit card generation • Send notifications	Parent View-only · child's data • Daily attendance + alerts • Result cards + admit card • Daily digest in-app & email	Student View-only · own data • Attendance history • Marks, report cards • Exam schedule + admit card

Access control is enforced at the API layer. Parents and students can only ever see their own (or their child's) data. Every privileged action — marks entry, publishing results, deleting students — is permission-checked on the server.

4. Key Workflows

From a teacher's tap to a parent's inbox.

Daily attendance → parent alert



Examination lifecycle

Step	Who	Action
1	Office Staff	Create exam period (Midterm / Pre-board / Board / Annual) with start & end dates.
2	Office Staff	Define exam schedule per subject, time slot, hall — system flags clashes.
3	Office Staff	Publish schedule; admit cards auto-available to every student.
4	Subject Teacher	Enter marks (per-student or bulk CSV).
5	Admin / Principal	Verify and publish results on the scheduled date.
6	Parent & Student	View report card online; download / print ICSC-compliant PDF.

5. Security, Privacy & Compliance

Designed with student data protection in mind.

AUTHENTICATION

JWT + secure cookies

Passwords hashed with bcrypt. Sessions delivered as Secure + HttpOnly cookies. 12-hour access tokens, 7-day refresh.

BRUTE-FORCE PROTECTION

Auto-logout after 5 attempts

Per-IP and per-account counters. 15-minute logout. Tracks real client IP behind reverse proxies.

AUTHORIZATION

Role-based access (RBAC)

Six roles, each with explicit allow-lists at every endpoint. Parents and students never see another student's data.

TRANSPORT

HTTPS end-to-end

Let's Encrypt certificates auto-renew. All traffic encrypted between browser and server.

DATA OWNERSHIP

On-premise / self-hosted

Database and storage live on your infrastructure. No third-party processor sees student records by default.

AUDITABILITY

Action history

Every mark entry, attendance update, and publish action is timestamped and attributed to a user account.

Compliant with common data-privacy principles (data minimisation, role separation, encrypted transport). The stack is ready for additional controls (audit-log export, IP whitelisting, MFA).

6. Integrations & Notifications

Channels

Channel	Use case	Status
In-app notifications	All users — appears in the bell icon	Live
Email (SendGrid)	Daily digest, low-attendance, exam announcements, result release	Pluggable — set API key
SMS (Twilio)	High-priority alerts for low-internet households	Optional add-on
WhatsApp	Broadcast announcements	Optional add-on

Daily parent digest (8 AM cron)

- Per-child attendance % over the last 30 days + warning if below 75%.
- Upcoming exams in the child's class (next 7 days).
- Recently released results.
- Today's school events and holidays.

The digest job is idempotent — it won't double-send on the same day — and can be triggered manually by the Principal at any time.

7. Dashboards & Reporting

Live insights, exportable for board meetings and compliance.

ADMIN DASHBOARD

Pulse of the school

Total students, average attendance, students below 75%, exams scheduled, next exam, results pending. 14-day attendance trend chart.

TEACHER DASHBOARD

Class snapshot

Class-specific attendance trend, pending tasks, quick links to mark attendance and enter marks.

PARENT / STUDENT DASHBOARD

Child snapshot

Attendance %, present / absent days, subject-wise performance chart, upcoming exams.

EXPORTABLE REPORTS

Spreadsheet & PDF

Attendance report, ranked results report, exam schedule, ICSC report cards. One-click CSV export.

8. Deployment & Scalability

Recommended sizing

School size	Server	Storage	Notes
Up to 500 students	2 vCPU · 4 GB RAM	50 GB	Single Ubuntu VM, everything co-located.
500 – 2,000 students	4 vCPU · 8 GB RAM	100 GB	Separate disk for MongoDB volume.
2,000 – 10,000 students	8 vCPU · 16 GB RAM	250 GB	Move MongoDB to its own server; add backup.

Operational practices

- Automated nightly database backups (mongodump).
- HTTPS certificate auto-renewal (certbot timer).
- Application logs via systemd · journalctl.
- Zero-downtime upgrades — restart backend without affecting frontend.
- Disaster recovery: full restore from latest backup in under 30 minutes.

Roll-out plan

Phase	Duration	Outcome
Setup & data migration	1 – 2 weeks	Server provisioned · classes, sections, students imported.
Pilot with 2 classes	2 weeks	Teachers trained · feedback collected · minor tweaks.
School-wide rollout	2 weeks	All teachers onboarded · parent logins distributed.
Steady-state	Ongoing	Daily digests, quarterly report cards, annual board reports.

9. Roadmap

What's next, beyond the launch.

QUARTER 1

Live email + SMS

Plug in SendGrid (and optionally Twilio) keys to enable real delivery. No code change required.

QUARTER 2

Weekly parent report

Sunday digest with attendance trend, latest grades, and a teacher-comment slot.

QUARTER 3

Mobile-first parent app

Native shell wrapping the existing web app, with push notifications.

QUARTER 4

Fee management module

Invoice generation, online payment integration, fee status on parent dashboard.

10. In Closing

A single platform — clear roles, calm UI, secure data.

ICSC Connect was built specifically for ICSC-affiliated schools that want a modern operational backbone without surrendering data ownership. It runs on infrastructure you already control, scales with your student body, and grows feature-by-feature alongside your needs.

Next conversation: walk through the live platform with your admin and teaching teams, agree on a pilot batch of classes, and lock in the rollout schedule.

Prepared for your school's IT & Academic Committee · 23 June 2026 · ICSC Connect School Management Platform